

Roles

In this chapter, we employ some of the results of our previous chapter to describe the modeling concept of roles.

Roles represent a fundamental notion for our conceptualization of reality. This notion has received much attention both in philosophical investigation (van Leeuwen, 1991; Wiggins, 2001; Loebe, 2003; Masolo et al., 2004, 2005) and in the conceptual modeling literature in topics such as object-oriented modeling (Bock & Odell, 1998), agent-oriented modeling (Odell & Parunak & Fliescher, 2003; Odell & Nodine & Levy, 2005), and organizational modeling (Dignum, 2004), among others. In a comprehensive study on this topic, Friedrich Steimann (2000b) defends that the role concept naturally complements those of *objects* and *relationships*, standing on the same level of importance. However, Steimann also recognizes that *“the role concept, although equally fundamental, has long not received the widespread attention it deserved”*, and that *“although there appears to be a general awareness that roles are an important modelling concept, until now no consensus has been reached as to how roles should be represented or integrated into the established modeling frameworks”* (ibid., p.84). The last statement can be verified by inspecting the diversity and incompatibility of the several conceptualizations of roles currently co-existing in the literature (Wieringa & Spruit & de Jong, 1995; Loebe, 2003; Steimann, 2000a,b, 2001; Bock & Odell, 1998).

Recently, not only has the interest in roles grown continuously, but also has the interest in finding a common ground on which the different notions of roles can be judged and reconciled. In fact, a significant part of the work presented in this chapter originated in a material that we prepared for an invited presentation in a workshop dedicated to discussing different conceptualizations of this concept (see Guizzardi, 2004, but also Masolo et al., 2005).

This chapter also exemplifies the usefulness of the categories proposed in our foundational ontology. Firstly, by employing the categories and

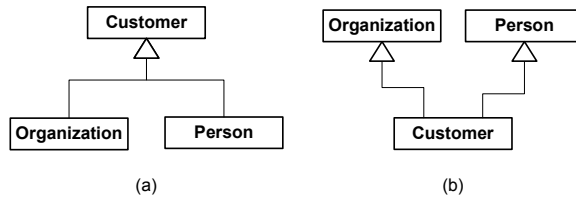
postulates of the theory of universals constructed in chapter 4, we propose an *ontological design pattern* that captures a solution to a recurrent and much discussed problem in role modeling (section 7.1). Moreover, with some definitions offered in chapter 5, we investigate the link between some of the formal meta-properties defined for part-whole relations and those meta-properties by which roles are characterized (section 7.2). Furthermore, by borrowing some results from chapters 4 and 6, we manage to harmonize some different conceptions of roles that are used in the literature (section 7.3). Finally, by building on an existing theory of transitivity of linguistic functional parthood relations, and on the role related notion of *qua individuals* discussed in section 7.3, we propose a number of visual patterns that can be used as a methodological support for the identification of the scope of transitivity for the most common type of part-whole relations in conceptual modeling (section 7.4).

Section 7.5 concludes the chapter by presenting some final considerations.

7.1 An Ontological Design Pattern for Role Modeling

Figure 7.1 below exemplifies a recurrent problematic case in the literature of role modeling (Steinman, 2001), which is termed here the problem of role with multiple disjoint *allowed types* (Bock & Odell, 1998). Suppose a conceptualization in which the social concept of Customer should be represented. A Customer, in this conceptualization, is assumed to be a role played by an entity in the context of *purchasing* relation to a Supplier. Moreover, in this conceptualization, not only persons but also organizations can be customers. Notice that, in this case, customer is indeed a role according to the definition of chapter 4, i.e., it is anti-rigid and relationally dependent universal. The problem is how to model the relationship between the role type Customer and its allowed types Person and Organization? This problem is also mentioned in (van Belle, 1999): “*how would one model the customer entity conceptually? The Customer as a supertype of Organisation and Person? The Customer as a subtype of Organisation and Person? The Customer as a relationship between or Organisation and (Organisation or Person)?*” The first two alternatives are presented in figures 7.1.a and 7.1.b, respectively.

Figure 7-1 Problems with modeling roles with multiple allowed types



In the model of figure 7.1.a, the role *Customer* is defined as a supertype of *Person* and *Organization*. This modeling choice violates postulate 4.3 (see chapter 4) and produces an ontologically incorrect conceptual model. Firstly, not all persons are customers, i.e., it is not the case that the extension of *Person* is necessarily included in the extension of *Customer*. Moreover, an instance of *Person* is not necessarily a *Customer*. Both arguments are equally valid for *Organization*.

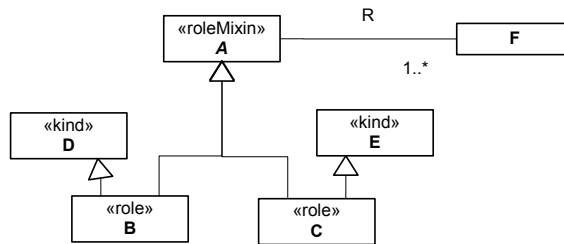
In a series of papers (Steimann, 2000a,b, 2001), Steimann discusses the difficulties in specifying admissible types for Roles that can be filled by instances of disjoint types. As a conclusion, the author claims that the solution to this problem lies in the separation of role and type hierarchies, which would lead to a radical and counterintuitive revision to the metamodel of most current conceptual modeling languages, or to put in UML terms, a counterintuitive *heavyweight extension* to the language (OMG, 2003b). In the sequel we show that this claim is not warranted. Moreover, we propose a *design pattern* based on the profile introduced in chapter 4 that can be used as an ontologically correct solution to this recurrent problem. Finally, this solution has a smaller impact to conceptual modeling metamodels than the one proposed by the author, since it does not demand radical heavyweight extensions to the language.

In the example above, *Customer* has in its extension individuals that belong to different kinds and, thus, that obey different principles of identity. *Customer* is hence a dispersive or transsortal type (a non-sortal) and, by definition, cannot supply a principle of identity for its instances. Since an (determinate) individual must obey one and only one principle of identity (postulates 4.1 and 4.2 in chapter 4), every instance of *Customer* must be an instance of one of its subtypes (forming a partition) that carry that principle of identity. For example, we can define the sortals *PrivateCustomer* and *CorporateCustomer* as subtypes of *Customer*. These sortals, in turn, carry the (incompatible) principles of identity supplied by the kinds *Person* and *Organization*, respectively. In summary, if *x* is a *Customer* (abstract class) then *x* must be an instance of exactly one of its subtypes (e.g., *PrivateCustomer*) that carries the principle of identity supplied by an appropriate substance sortal (e.g., *Person*).

Figure 7.2 shows how this solution can be incorporated in a conceptual modeling design pattern. In this picture the abstract class *A* is the *role mixin*

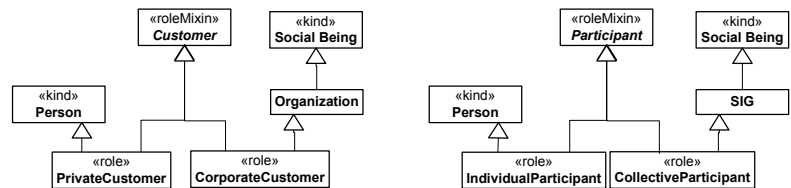
that covers different role types (e.g., Customer, Participant). Classes B and C are the disjoint subclasses of A that can have direct instances, representing the sortal roles that carry the principles of identity that govern the individuals that fall in their extension. Classes D and E are the ultimate substance sortals (kinds) that supply the principles of identity carried by B and C, respectively. The association *r* represents the common specialization condition of B and C, which is represented in A. Finally, class F represents a type that A is *relationally dependent* of.

Figure 7-2 An ontological design pattern for the problem of specifying roles with multiple disjoint allowed types



An application of this pattern is illustrated in figure 7.3 in which it is used to produce ontologically correct versions of two stereotypical models of this situation (one of them being the model depicted in figure 7.1). In both cases, the universal the *role mix*in depends of, and the association representing the specialization condition are omitted for the sake of brevity.

Figure 7-3 Ontologically correct models of roles with disjoint allowed types obtained by the application of the design pattern of figure 7.2



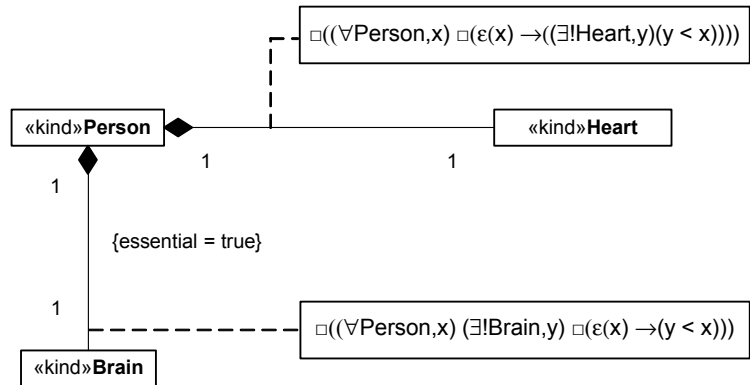
7.2 Parts of Roles

In chapter 5, we have presented a division of parthood relations w.r.t. ontological dependence containing three possible subtypes:

- *essential parts*: characterized by existential dependence from the whole to a part;
- *mandatory parts*: characterized by generic constant dependence from the whole to the universal a part instantiates;
- *contingent parts*.

Let us take the human body as an example. The relation between an individual Human Body and an individual Human Brain is an example of (i). In contrast, the relation between a Human Body and a Human Heart is an example of (ii). In the former case, a Body x depends specifically on the existence of a particular Brain y , i.e., for every instance of Human Body x there is a unique Brain y such that in every world that x exists, y is part of x . In the latter case, contrariwise, the Body x depends on the existence of *any* instance of the Heart universal, not on a specific one. In other words, for every Human Body x and in every world that x exists, there is a Human Heart y that, in that world, is part of x . These two situations are depicted in figure 7.4 together with their corresponding modal logics formalizations. For the sake of simplicity, we formalize in this case only the axioms w.r.t. the relation from the whole to the part. All other axioms are omitted. Moreover, one should remember that we are dealing only with non-trivial entities here, i.e., we dispense with entities that are necessarily existent or necessarily non-existent.

Figure 7-4
Representation of
essential and mandatory
parthood in a conceptual
specification of the
human body

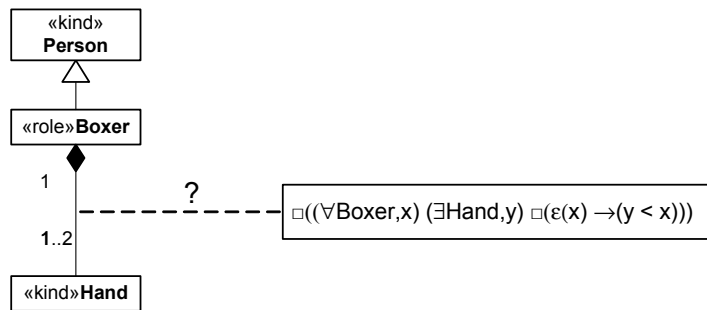


In all examples used in chapter 5, the universals representing wholes are *kind universals* (see chapter 4). Let us now investigate how these different types of necessary parthood relations can be used to characterize other non-rigid universals, such as *Roles*. Suppose, for instance, the situation depicted in figure 7.5. The figure illustrates the relation between a Boxer and one of his hands. What the picture attempts at representing is the statement that “every boxer must have a hand”. This relation is certainly not one of mandatory parthood, since it is not the case that a Boxer depends generically on the universal hand but specifically on one particular hand⁶³. It thus appears to be the case that this relation is one of essential parthood.

⁶³ We are here not considering the possibility of hand transplants. Once more, the point of the argumentation is not the specific example.

However, this is not true either. If a hand were to be considered an essential part of a particular boxer then the corresponding formula represented in figure 7.5 should be valid. To show that this is not the case, suppose the following: let John be a boxer in world w and let x be John's hand in w . What the formula in figure 7.5 states is that in every world w' in which John exists, x must be part of John in w' . This formula is clearly falsifiable. One just have to imagine a world w'' , in which John exists without being a boxer and without having x as his hand (supposed that x has been tragically amputated in w''). This problem arises from the ambiguity of the word “must” in “every boxer must have a hand”. Intuitively, the situation that this model intended to express is the valid statement that “For every Person x , there is a hand y , such that *in every world that x is a Boxer, y is a hand of x* ”.

Figure 7-5 Problems in the representation of specifically dependent parts for anti-rigid universals



In the example of figure 7.5, Boxer cannot have essential properties and, in particular, cannot have essential parts, since it is an anti-rigid universal. In other words, if “to be a boxer” is considered as a property, it is not an essential property itself of any individual.

However, this situation can be understood in terms of the philosophical distinction between *de re* and *de dicto* modality. Take the following two sentences:

- (i) The queen of the Netherlands is necessarily queen
- (ii) The number of planets in the solar system is necessarily odd

In the *de re* reading, the first sentence expresses that a certain individual (Beatrix) is necessarily queen. This is clearly false, since we can conceive a different world in which Beatrix decides to abdicate the throne. However, in the *de dicto* reading the sentence simply expresses that it is necessarily true that in any circumstance whoever is the Dutch queen is a queen. The second sentence works in the converse manner. In the *de re* reading the sentence (ii) expresses that a certain number (9) is necessarily odd. This is

indeed necessarily true. The *de dicto* reading of the sentence however is false. It is not necessarily the case that the number of planets in the solar system is odd. We can imagine a counterfactual situation in which the solar system has, for instance, 8 or 10 planets. The Latin expressions *de re* represents a modality which refers to a property of the thing itself (*res*), whereas *de dicto* represents a modality that refers to an expression (*dictum*). This is made explicit in the logical rendering of the possible readings of these two expressions:

(iii-a) *de re* (false): $\forall x \text{ QueenOfTheNetherlands}(x) \rightarrow \Box(\text{Queen}(x))$

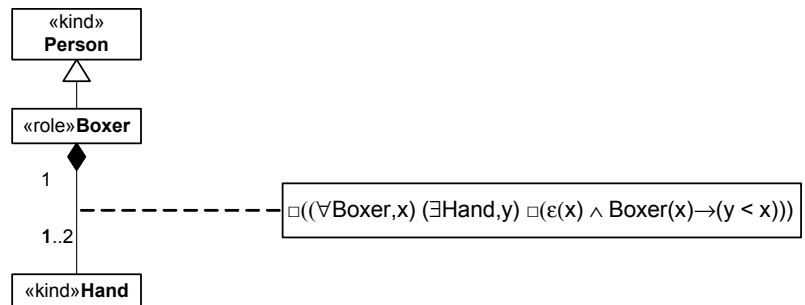
(iii-b) *de dicto* (true): $\Box(\forall x \text{ QueenOfTheNetherlands}(x) \rightarrow \text{Queen}(x))$

(iv-a) *de re* (true): $\forall x \text{ NumberOfPlanets}(x) \rightarrow \Box(\text{Odd}(x))$

(iv-b) *de dicto* (false): $\Box(\forall x \text{ NumberOfPlanets}(x) \rightarrow \text{Odd}(x))$

Take now the expression “every boxer has necessarily a hand”. Once more, this expression is true only in one of the readings, namely, the *de dicto* reading. Whilst it is the case that the expression “In any circumstance, whoever is boxer has at least one hand” is necessarily true, it is false that “If someone is a boxer than he has at least a hand in every possible circumstance”. Figure 7.6, expresses a correct representation of this situation in the *de dicto* modality.

Figure 7-6 Correct representation of specifically dependent parts of anti-rigid universals



We now have expressed three different types of dependency relations between wholes and parts:

- specific dependence with *de re* modality;
- generic dependence with *de re* modality;
- specific dependence with *de dicto* modality.

The remaining option is, of course, conceivable, i.e., generic dependence with *de dicto* modality. This situation can be captured by the following

formula (v), in which A represents the (anti-rigid) whole and B represents the part. In this formula, the predicate B is used as what we term here a *guard predicate*. Intuitively, this predicate “selects” those worlds, in which the parthood relation must hold. The same holds for the predicate Boxer in figure 7.6.

$$(v) \quad \Box(\forall A, x \Box(\mathcal{E}(x) \wedge A(x) \rightarrow \exists y B(y) \wedge (y < x)))$$

We have seen that essential properties, i.e., specific dependence expressed in terms of the *de re* modality, can only be expressed for rigid universals. For anti-rigid universals (e.g. roles), only the corresponding *de dicto* modality can be applied. Nonetheless, it is also true that for every *de re* statement regarding an individual x , we can express a corresponding *de dicto* one, by using as guard predicate the substance sortal that x instantiates. For instance, if it is true that “The number of planets in the solar system (9) is essentially odd” then it is also true that “In any circumstance, if 9 is a number then 9 is odd”. We therefore could rephrase the formulas in figure 7.4 as follows:

$$(vi) \quad \Box((\forall \text{Person}, x)(\exists! \text{Heart}, y) \Box(\mathcal{E}(x) \wedge \text{person}(x) \rightarrow (y < x)))$$

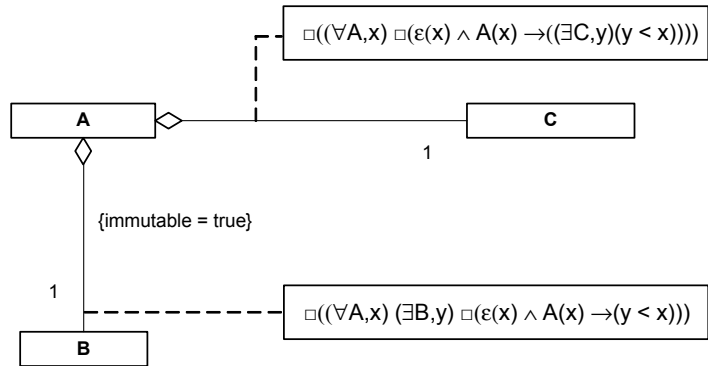
$$(vii) \quad \Box((\forall \text{Person}, x) \Box(\mathcal{E}(x) \wedge \text{person}(x) \rightarrow (\exists! \text{Heart}, y)(y < x)))$$

Since Person is a substance sortal (rigid universal), everything that is person is necessarily a person (see chapter 4). In other words, the predicate *person* is modally constant, and for every object selected by the universal quantifier, *person* must be true for this object in every possible world. Consequently, (vi) and (vii) are logically equivalent to their counterparts in figure 7.4.

In order to achieve a uniform axiomatization, we therefore propose the following formula schemas depicted in figure 7.7, which must hold irrespective of the whole universals being rigid or anti-rigid sortals. If the universal A is rigid then $A(x)$ is necessarily true (if true) and the antecedent $(\mathcal{E}(x) \wedge A(x))$ can be expressed only by $(\mathcal{E}(x))$. In this case, the B’s are truly essential parts of A’s.

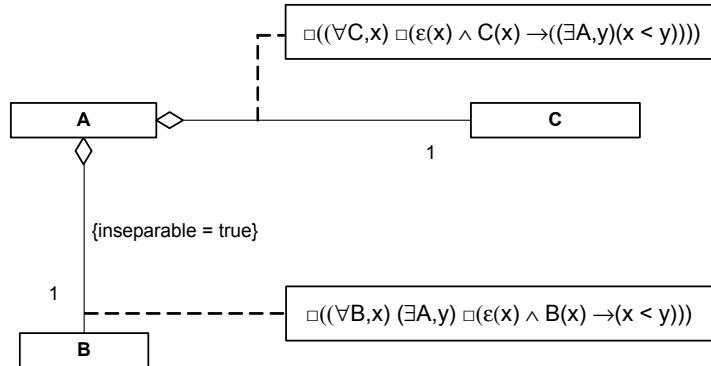
From now on, we refrain from using the term *essential parts* for the cases in which a mere *de dicto* modality is expressed. Therefore, for the case of *specific dependence* from instances of roles (and role mixins) to theirs part we adopt the term *immutable part* instead. Of course, every essential part is also immutable.

Figure 7-7 General representation for Immutable and Mandatory parts



Generalization axioms analogous to those in figure 7.7 can be produced for the case of inseparable and mandatory wholes. Figure 7.8 depicts a representation of inseparable parts and mandatory wholes, in which guard predicates are included to produce generalizations of the axioms in definitions 6.14 and 6.15 that are suitable for the cases of both rigid and anti-rigid universals.

Figure 7-8 General representation for Inseparable Parts and Mandatory Wholes

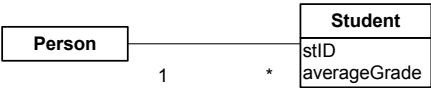


7.3 Harmonizing different notions of Roles in Conceptual Modeling

In chapter 4, we have discussed the related work of (Wieringa, de Jong and Spruit, 1995), which discusses the need for elaborating on the distinctions among the types of universals used in conceptual modeling. Wieringa and colleagues propose three classifier categories: *static classes*, *dynamic classes* and *roles*. The first two of these correspond to our categories of kinds (and subkinds) and phases, respectively. However, differently from our proposal,

in their approach a role universal is not a phased-sortal. Conversely, their roles are *rigid* universals whose instances are said to be *played by* instances of ordinary (static and dynamic) types. The *played by* relation (also termed *inheritance by delegation* by the authors) between a role instance *r* and an object *o* implies that *r* is *existentially dependent* on *o*. This means that *r* can only be played by *o*, and that *r* can only exist when played by *o*. However, in contrast, *o* can possibly be associated via the *play by* relation to many instances of the role class (and to many different role classes). Moreover, role universals are responsible for supplying a principle of identity for its instances, which is different from the one supplied by the universals instantiated by their players. Figure 7.9 depicts an example of ordinary and role universals according to the authors.

Figure 7-9 Example with Role and Role player universals



An inspection of the role literature shows, however, that most authors conceive role universals in a way which is akin to the notion proposed in chapter 4, i.e., as substantial universals. This includes authors both in philosophy (Wiggins, 2001; van Leeuwen, 1991) and in conceptual modeling in computer science (Bock & Odell, 1998; Essink & Erhart, 1991; Jungclaus et al., 1991; Sowa, 1984, 1988). Moreover, several authors share the view that the identity of a role instance is supplied by a universal subsuming the role type that it instantiates (i.e., the role’s allowed type) (Gottlob & Schrefl & Rock, 1996; Kristensen, 1995; Albano et al., 1993; Richardson & Schwartz, 1991). Finally, there are authors that explicitly share both views see (Guarino & Welty, 2004, 2002a,b, 2000; Steimann, 2001, 2000a,b). In fact, in an extensive study about the topic of roles in the conceptual modeling literature, (Steimann, 2000b) deems the approach of Wieringa and colleagues to be a singular case in which the identity of role instances is not supplied by a universal subsuming the role type they instantiate.

The motivation for such a view proposed by Wieringa and colleagues lies in a philosophical problem known as *The Counting Problem* (Gupta, 1980). Consider the following argument:

KLM served four thousand passengers in 2004
Every passenger is a person
Ergo, KLM served four thousand persons in 2004

Thus, as Wieringa et al. write: “if we count persons, we may count 1000, but if we count passengers, we may count 4000. The reason for this difference is that if we

count things we must identify those things, so that we can say which things are the same and which are different. But in order to identify them, we must classify them."

Although, we appreciate and share the view of connecting *counting with identity* and *identity with classification* (see chapter 4), we do not agree with the conclusion the authors make of this example, namely, that since person and passenger do not share a principle of counting then they must not share a principle of identity either. Since, as we discuss in chapter 4, a principle of identity can only be supplied by a rigid universal. This must be the foundation of the authors' conclusion that a role universal therefore must be a rigid universal.

Why do we think the conclusions made by the authors are not warranted? To start with, in line with (van Leeuwen, 1991), we defend that the counting problem is actually a fallacy. Take the argument posed by its defenders: "The person that boarded flight KL124 on April 22nd, 2004 is a different passenger from the person who boarded flight KL256 on November 19th, 2004, but the two passengers are the same person". We do not agree that it can be correctly said that *the two passengers are the same person*, or, alternatively, that *a single person is distinct passengers* (at different times), if we are truthful to our commonsense use of the *common noun* passenger. However, let us suppose that this is the case, i.e., the person and passenger obey different principles of identity. In this situation, the second premise of the argument is no longer valid, i.e., one cannot say anymore that *every passenger is a person* in a reading in which the copula "is" is interpreted as a relation of identity. This is because, due to Leibniz Law, identity holds necessarily and that identical things necessarily share all their properties (principle of indiscernability of the identicals). As a consequence, we would arrive at the invalid conclusion that every person has all the properties of a passenger in all situations he/she is a person. Moreover, since identity is an equivalence relation, we would have that

"passenger *x* on flight KL124" is identical to person *y*

"passenger *z* on flight KL256" is identical to person *y*

Ergo, "passenger *x* on flight KL124" is identical to "passenger *z* on flight KL256"

This conclusion contradicts the initial premise that the two passengers were different. Therefore, if we have the second premise interpreted in the strong reading, one must conclude that passenger carries the same principle of identity as person and, hence, that "passenger *x* on flight KL124" and "passenger *z* on flight KL256" are indeed numerically the same.

In this case, though, the first premise ceases to be true, i.e., one can no longer say that "KLM served two million passengers in 2004". We must conclude then that the second premise should have a weaker reading in

which the copula does not represent a relation of identity but one of *coincidence*. But, if this interpretation is taken the whole argument is clearly invalid, since the conclusion cannot be expected to follow from the premises.

In summary, the conclusion that different principles of identity must be supplied by role types and the types instantiated by their players cannot follow from this argument. Furthermore, as pointed out by (Putnam, 1994), if the principle of identity of passengers were not supplied by a unique substantial sortal subsuming passenger (see chapter 4), and if passenger were a rigid classifier whose identity would be tied to the context of particular flight, we would not be able to make sense of sentences such as “The passenger came on flight KL109 and she continued on flight 311”.

Despite disagreeing with the conclusions, we think there is an important truth highlighted by the argument of Wieringa and colleagues. If not instances of passengers, what is one counting when stating that “KLM served two million passengers in 2004”? Let us analyze the concept of role proposed by Wieringa et al.:

1. a role universal is a rigid classifier;
2. role instances are (one-sidedly) existentially dependent of a unique object, which is said to *play* the role;
3. objects *play* these roles only contingently, i.e., the *play* relation is only a contingent relation for the player. As a consequence, ceasing to play the role does not alter the identity of the player object.

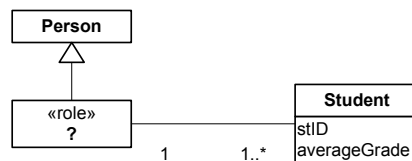
A recent work that has a concept of role similar to the one of Wieringa et al. is (Loebe, 2003). However, Loebe’s roles are not only *existentially dependent* on their players, but they also depend on the existence of another entity (distinct from their players), in the way, for instance, that *being a student* depends on the existence of a education institution, or *being a husband* depends on the existence of a wife, *being an employee* depends on the existence of an employer, etc. This feature of roles is recognized in our analysis in chapter 4. In fact, it is generally accepted in the literature that roles only exist in a certain context, or in the scope of a certain relation (Masolo et al., 2004, 2005; Guarino & Welty, 2004, 2002a,b, 2000; Steimann, 2001, 2000a,b; Bock & Odell, 1998; Chu & Zhang, 1997; Elmasri & Weeldreyer & Hevner, 1985; Sowa, 1984). Thus, Loebe’s notion of roles agrees with that of Wieringa et al. in the points (a), (b) and (c) above, but it also characterizes role instances as existentially dependent on each other.

The concept of role in Wieringa et al. and Loebe is equivalent to our notion of *qua individuals* discussed in chapter 6, since we can interpret their *play by* relation as a sort of inherence. Both relations represent a one-side

monadic existential dependence relation. Thus we can say that, like their notion of Roles, our *qua* individuals (special types of externally dependent modes) are: instances of a rigid classifier (a); one-side existentially dependent of objects, which are related to their “players” via a contingent sort of existential dependence relation (b). Furthermore, a *qua* individual is a complex of externally dependent modes (e.g., in figure 7.9, *student id*, *average grade*⁶⁴), which, by definition, depends also on the existence of another object extrinsic to its bearer (player). Thus, as in Loebe’s concept of roles, asides from the inherence (play) relationship with its bearer (player), our *qua* individuals stand in parthood relationship with a unique relator in the scope of a material relation. Since relators consist of at least two distinct *qua* individuals, we conclude that the *qua* individuals composing a relator are existentially dependent on each other.

Now, how can we relate this notion of roles as *qua* individuals with the one proposed in chapter 4? Let us revisit the example depicted in figure 7.9 above. To start with, a point that can be argued against this model is the representation of optional cardinality constraints. In fact, since no restriction is defined for the allowed type of a role classifier, optional cardinalities must be represented in both Wieringa’s and Loebe’s approaches. As argued in, for instance, (Weber, 1997; Wand & Storey & Weber, 1999), from an ontological standpoint, there is no such a thing as an optional property and, hence, the representation of optional cardinality leads to unsound models (in the technical sense of chapter 2), with undesirable consequences in terms of clarity. Moreover, as empirically demonstrated in (Bodard et al., 2001), conceptual models without optional properties lead to better performance in problem-solving tasks that require a deeper-level understanding of the represented domain. To put it simply, not all persons bear a student moment, but only those persons that, for example, are enrolled in an educational institution. We can then define a restriction of the universal Person, whose instances are exactly those individuals that bear a student moment, i.e., that are enrolled in an educational institution (see figure 7.10).

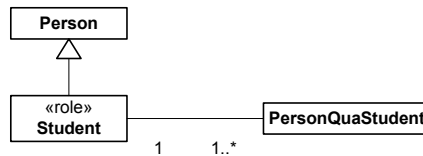
Figure 7-10 A Role universal, its allowed type and an exemplification relation to a *qua* individual universal



⁶⁴ To see that, for example, having a particular *student id* is an externally dependent moment, the reader should imagine a person that is registered in different departments of a university, having a different student id for each department.

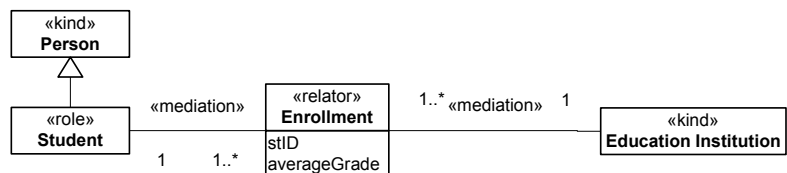
Now, the role universal in figure 7.10 is exactly what we mean by a role in chapter 4 and it is the one idea of role that accurately corresponds to the commonsense use of roles in ordinary language. For this reason, we propose to use the role name for the role universal and to create a new name for the *qua individual* universal (see figure 7.11). This choice is also supported by Aristotle's analysis of substantial and moment universals: student as a universal is *said of subject* (*de subjecto dici*) not exemplified in a subject (*in subjecto est*). That is why the general term Student (Passenger, Employee, etc.) belongs to the grammatical category of *count nouns*, not to the category of *adjectives*⁶⁵.

Figure 7-11 A Role universal, its allowed type and an exemplification relation to a *qua individual* universal (revised version)



Although an improvement of figure 7.10, figure 7.11 is still incomplete in the sense that it does not express the additional dependence relation that a *qua individual* has with other objects extrinsic to its bearer (see definition of *qua individual* in chapter 6). This problem is solved in figure 7.12, in which relators (as aggregates of *qua individuals*) are represented explicitly and in which the externally dependent moments of a *qua individual* are represented as *resultant moments*⁶⁶ of the relator. In this figure, the associations between Student and Enrollment and between Education Institution and Enrollment stand for formal relations of mediation (see chapter 6).

Figure 7-12 A role universal, its allowed type and its associated relator universal



Now, let us return to the “counting problem” previously discussed:

⁶⁵ Etymologically the English word *noun* comes from the latin word *substantivus*, meaning expressing *substance*. The original form is still preserved in latin languages such as Portuguese (*substantivo*) and Italian (*sostantivo*), as well as in the English word *substantive*, which is a less familiar synonym for noun. Conversely, one of the meanings of *adjective* in English is “not standing by itself, dependent” (Merriam-Webster, 2004).

⁶⁶ Resultant properties of an object are properties that a whole inherits from one of its parts (Bunge, 1977, p.97).

500 students graduated from the University of Twente in 2004
 Every student is a person
 Ergo, 500 persons graduated from the University of Twente in 2004

In this argument, if the first premise is true than the word *student* refers to the mode *Person qua student*. The counting of these entities in a given situation is equal to the cardinality of the extension of the *PersonQuaStudent* universal in figure 7.11 (i.e., $\#_{ext}(\text{PersonQuaStudent})$) or the cardinality of the extension of the *Enrollment* classifier in figure 7.12 (i.e., $\#_{ext}(\text{Enrollment})$), since there is always a 1-1 correspondence between relators and their composing qua individuals. However, if this interpretation for student is assumed, the second premise is simply false, since the relation between a student and a person would be one of inherence, not one of identity. Alternatively, if the word *student* is interpreted (in the more natural way) as in figure 7.11, then the counting of students is equal to the cardinality of the extension of the *Student* universal in this figure (i.e., $\#_{ext}(\text{Student})$). Though, in this case, premise one is not necessarily true.

In both cases, the alleged “counting problem” disappears. Nonetheless, with the model of figure 7.12 we are still able to represent for both kinds of entities (roles and qua individuals) and their respective counting in an unambiguous manner. Additionally, this solution is able to make explicit and harmonize the two diverse senses of Role which have been used in the conceptual modeling literature. Finally, as we demonstrate in section 6.3.3, there are other important conceptual modeling benefits of this type of representation proposed.

In the same manner that qua individuals are employed to characterize roles in this section, one can think about other types of qua individuals that characterize other types of sortal universals. In this spirit, we can think about a *phase qua-Individual* as (potentially complex) *intrinsic* moments that a substantial bears when in a given phase, or a *substantial qua-Individual*⁶⁷ as a potential complex of essential moments, which thus inhere necessarily in an individual. Any attribution (non-sortal) is, in fact, founded on a characterization relation to what can be conceived as some sort of qua individual universal (in the limit case any moment can be thought as a simple qua individual). However, in this thesis, when using the term *qua individual* we mean a *relational qua-Individual*, i.e., a special type of externally

⁶⁷ In the Aristotelian tradition there is always a unique universal that answers the question of “what a thing is?” According to Aristotle, seeing an object as an instance of this universal is equivalent to seeing the *individual qua itself* (van Leeuwen, 1991). This universal is named here a kind (or substance sortal).

dependent mode that a substantial has in the context of a certain material relationship.

Finally, we can refine the characteristic *relational dependence* defined for roles in chapter 4 by explicitly relating the notions of role and qua individuals discussed in this section. As discussed in section 4.1, roles are defined as specializations of a substance sortal according to a relational specialization condition ϕ . Here, this specialization condition can be further analyzed as a material relation being derived from a certain relator universal U_R (see definition 6.17). Consequently, we can state that a role universal bears always a mediation relation to a relator universal.

$$(1). \forall x \text{ Role}(x) \rightarrow \exists y \text{ RelatorUniversal}(y) \wedge \text{mediation}(x,y)$$

Or alternatively, that a role universal is always characterized by a qua individual universal:

$$(2). \forall x \text{ Role}(x) \rightarrow \exists y \text{ QuaIndividualUniversal}(y) \wedge \text{characterization}(x,y)$$

As a consequence of formula (1), we have that, in the UML profile defined in chapter 8 of this thesis, a UML class stereotyped as «role» must always be connected to an association end of a «mediation» relation (see section 8.2.2).

7.4 Qua Individuals and the problem of transitivity in mandatory parthood relations between functional complexes

As we have discussed in chapter 5, the parts of a complex object have in common the fact that they all possess a functional link with the complex. In other words, they all contribute to the functionality (or the behavior) of the complex. Moreover, parthood between complexes also represents a case of *functional dependence*. In section 5.6, we have addressed the case of *specific functional dependence* between complexes, exemplified, in a given conceptualization, by the relation between an individual person and the specific brain of that person.

In figure 7.13, we represent a parthood relation between two complexes. However, the functional dependence that is implied by the parthood relation in this example is not one of specific dependence. Conversely, this type of relationship between the universals Heart and Body

is what is named *Generic Functional Dependence* between two universals in (Vieu & Aurnague, 2005). This relationship can be defined as follows:

$$(3). \text{GFD}(X,Y) \equiv \Box(\forall x (x::X) \wedge F(x,X) \rightarrow \exists y \neg(y = x) \wedge (y::Y) \wedge F(y,Y))$$

Figure 7-13 A
mandatory parthood
relationship between two
Complex Object
Universals

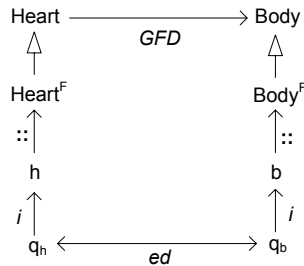


The predicate $F(x,X)$ in formula (3) has the meaning *x functions as an X*. In Vieu and Aurnague's theory, it is not necessary for an X that it functions as an X . So for instance, although Engine is a rigid concept, it is not the case that in every circumstance an engine functions as an engine. We thus can think of a universal X^F which is a restriction of X according to the restriction condition expressed by the predicate $F(x,X)$, so that every X^F is a X functioning as a X . We name the universal X^F a functional restriction of X . Notice that X^F in this case is an anti-rigid universal which can be characterized by the qua individual q_x . This qua individual, in turn, stands for the moments bearing in an X 's while functioning as such, or the particular behaviour of an X while functioning as an X . For instance, an engine x can have the property of emitting a certain number of decibels or being able to perform certain tasks only when functioning as an engine.

In figure 7.14, we can create restrictions of the universals Heart and Body to the universals Heart^F (*FunctioningHeart*) and Body^F (*FunctioningBody*). In this picture, the arrow with the hollow head represents subsumption. The symbols $::$, i and ed represent instantiation, inherence and existential dependence, respectively. Whenever a heart functions as such, i.e., whenever it instantiates the universal *FunctioningHeart*, there is a qua individual q_h that inheres in it. *Mutatis Mutandis*, the same goes to Body and *FunctioningBody* in this picture. As represented in this picture, the qua individuals q_h and q_b are existentially dependent on each other. In this case, $ed(q_h, q_b)$ can be interpreted as "the heart functioning behavior existentially depends on the body functioning behavior". In this model the converse also holds, i.e., that $ed(q_b, q_h)$, or that "the body functioning behaviour existentially depends on the heart functioning behavior". Additionally, according to our model, a heart functioning q_h must inheres a heart h . Likewise, a body functioning q_b must inheres a body b . From this we have that whenever a heart h functions as a heart (i.e., $i(q_h, h)$) there must exist a body functioning behavior q_b (from $ed(q_h, q_b)$), which in turn, inheres a body b (i.e., $i(q_b, b)$). In other words, whenever a heart h functions as a heart, there must be a body b functioning as a body. Again, from the model of figure 7.14 we can derive the converse information, namely, that

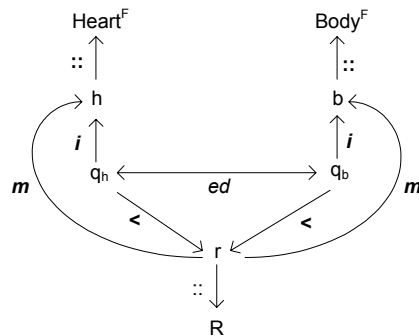
whenever a body b functions as a body, there must be a heart h functioning as a heart.

Figure 7-14
Representation of
universals with Generic
Functional Dependence
and their Functional
Restrictions



By definition of the relational qua individuals we are considering here (see chapter 6), q_h and q_b in figure 7.14 are externally dependent modes that share the same foundation and therefore can be said to compose a relator r that, in turn, can be said to mediate the instances of *FunctioningHeart* and *FunctioningBody*. This idea is depicted in figure 7.15. The symbols m and $<$ in this picture represent the mediation relation and the proper parthood relation, respectively.

Figure 7-15
Representation of the
relator instance
composed of two
functional qua
individuals



As discussed in chapter 6, the relator universal R of which the relator r in figure 7.15 is an instance, can be said to induce the material relation Φ_R between the universals *FunctioningHeart* and *FunctioningBody*. We shall define here the more general binary predicate $\Phi(x,y) \equiv \exists r m(r,x) \wedge m(r,y)$. In other words, $\Phi(x,y)$ holds iff there is a relator r which mediates these two individuals. More naturally, in this case, we can say that Φ hold of x and y of type X and Y iff x to function as an X is depends on y functioning as a Y , and vice-versa. Notice that the functional restriction *FunctioningHeart* (*FunctioningBody*) is indeed not only an anti-rigid universal but also a *relationally dependent* one and, consequently, it conforms to the formulas (1) and (2) previously discussed: a *FunctioningHeart* is a *Heart* functioning as a

Heart in relation to a Body functioning as a Body, and vice-versa. To put in different terms, these functional restrictions of universals are types of Roles.

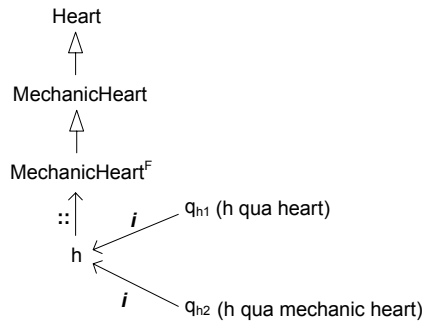
The predicate Φ to hold for instances of functional restrictions X^F and Y^F requires the presence of a relator r to mediate these instances. This requires that the functional qua individuals inhering in the mediated instances of X^F and Y^F share a genuine foundation. The formula (3) of generic functional dependence between X and Y can then be better expressed as:

$$(4). \text{GFD}(X,Y) \equiv \Box(\forall x (x::X) \wedge F(x,X) \rightarrow \exists y (y::Y) \wedge F(y,Y) \wedge \Phi(x,y))$$

Notice that the predicate Φ eliminates the possibility that formula (4) is trivially satisfied by its consequent being necessarily true. Another possibility for this formula being trivially satisfied is if its antecedent were to be necessarily false. This would imply that the $(\neg\Diamond\exists x (x::X) \wedge F(x,X))$ is true, which in turn implies that $(\neg\Diamond\exists x x::X^F)$. However, this amounts at stating that the universal X^F cannot be possibly instantiated. By adopting an Aristotelian theory of universals in chapter 6 and, therefore, rejecting the possibility of platonic universals, we require that every universal in the theory must be possibly instantiated. We conclude then that formula (4) cannot be trivially satisfied. Moreover, as discussed in chapter 6, a relator must mediate at least two distinct individuals. As a consequence, we have that $\Phi(x,y)$ implies $\neg(y = x)$, rendering this condition superfluous in the consequent of formula (4).

Suppose that the universal X is a specialization of another universal A . Then not only every X is an A but whenever an X functions as such it also functions as an A (Vieu & Aurnague, *ibid.*). For example, suppose that X and A are the universals *MechanicHeart* and *Heart*, respectively. Whenever a *MechanicHeart* functions as a *MechanicHeart*, it also functions as a *Heart*, or alternatively, whenever a *MechanicHeart* bears the behaviour (or properties) of a functioning *MechanicHeart*, then it also bears the properties of a functioning *Heart*. This is illustrated in figure 7.16.

Figure 7-16
Propagation of
Functioning to the
Supertype



We thus have that

$$(5). (F(x, X) \wedge \text{Subtype}(X, Y)) \rightarrow F(x, Y)$$

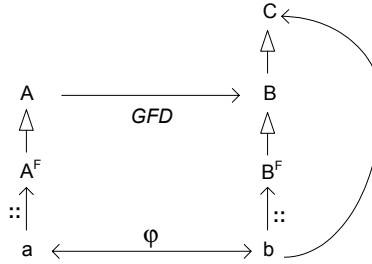
Moreover, we assume the following to hold:

$$(6). F(x, X) \rightarrow \mathcal{E}(x)$$

That is to say, if x is functioning as an X in a given world then x exists in that world. As previously mentioned, we assume that concrete individuals are not necessarily existents. Therefore, we conclude that an individual X cannot necessarily function as an X either. In other words, a functional restriction of a universal is indeed an anti-rigid universal.

Suppose the situation depicted in figure 7.17.

Figure 7-17
Propagation of General
Functional Dependence
to the Supertype

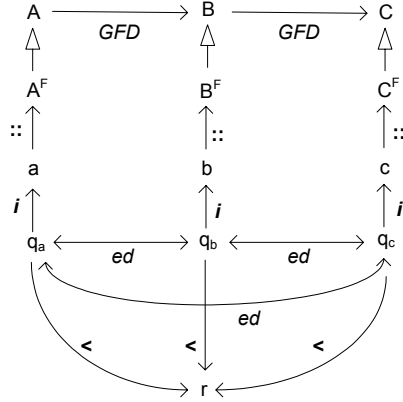


The universal A is generally functionally dependent on universal B . Thus, for every instance a of A that functions as such there is an instance b of B functioning as a B . Moreover, the predicate ϕ holds for a and b . Now, since b is also a C and, due to (5), b also functions as a C . Hence, we have that whenever an instance a of A functions as such there is an instance b of C that functions as a C . Since $\phi(a, b)$, we can derive that $GFD(A, C)$. Thus, we have that the following is always true:

$$(7). GFD(X, Y) \wedge \text{Subtype}(Y, Z) \rightarrow GFD(X, Z)$$

Now, suppose the situation depicted in figure 7.18.

Figure 7-18 Transitivity
of General Functional
Dependence



In this model, every instance a of A functioning as an A bears a particular q_a behaviour. The qua individual q_a is existentially dependent on the qua individual q_b , i.e., on the behaviour of a b functioning as a B . However, this model also represents that if b functions as a B (bears q_b) there is a c functioning as a C , i.e., bearing a C behavior q_c . Due to transitivity of existential dependence, we have that q_a is existential dependent also on q_c . Additionally, q_a and q_b share the same foundation and so do q_b and q_c . Thus, q_a and q_c also must share the same foundation. In other words, whatever is responsible for creating q_a and q_b must also be responsible for creating q_c . By definition (see chapter 6), a relator is an aggregation of qua individuals that share the same foundation. We can then define a relator r which consists of q_a , q_b and q_c . Consequently, we have that $\Phi(a,b)$, $\Phi(b,c)$ and $\Phi(a,c)$. Now, we have that for every instance a of A functioning as an A , there is an instance of c functioning as a C . Since $\Phi(a,c)$, we then have that $GFD(A,C)$. This argument shows that the following is always true:

$$(8). GFD(X,Y) \wedge GFD(Y,Z) \rightarrow GFD(X,Z)$$

Although formula (4) defines the notion of general dependence, we need in addition to establish that a functional dependence link holds precisely between two individual entities x and y :

$$(9). IFD(x,X,y,Y) \equiv GFD(X,Y) \wedge x::X \wedge y::Y \wedge (F(x,X) \rightarrow F(y,Y))$$

This predicate termed *individual functional dependence* states that if an individual $x::X$ is *individually functionally dependent* of another individual $y::Y$ then: (i) there is a generic functional dependence between their types; (ii) x

and y are classified as those given types in that world; (iii) for x to function as a X in that world, then y must function as a Y .

An example of individual functional dependence is one between a particular heart h and a particular body b in figures 7.14 and 7.15. As previously discussed, there is a generic functional dependence between the universals Heart and Body, and if in a given circumstance a heart h functions as a heart there is a body b that functions as a body in that circumstance.

Let us now return to the example of figure 7.13 of a mandatory parthood relation between the universals Heart and Body. In this model, a particular heart h is not only functionally dependent of a body b in a given world, but h is also part of b . This type of the parthood relation is termed in (Vieu & Aurnague, 2005) *direct functional parthood of type 1*:

Definition 7.1 (Direct Functional Part of type 1): An individual x instance of X is a direct functional part of type 1 of an individual y of type Y (symbolized as $d_1(x, X, y, Y)$) iff x is a part of y and x is individually functionally dependent of y . Formally, $d_1(x, X, y, Y) \equiv ((x < y) \wedge \text{IFD}(x, X, y, Y))$.

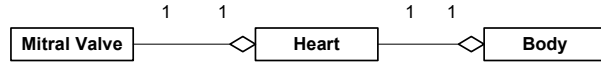
■

Examples of d_1 include cuff-sleeve, stem-plant, carburetor-engine, finger-hand, hand-arm, arm-body, hand-body, heart-body, heart-circulatory system.

In conformance with the findings of (Vieu & Aurnague, 2005), we propose that a mandatory parthood relation between two *functional complexes*⁶⁸ (such as the one depicted in figure 7.13) should be interpreted as a case of *direct functional parthood*. In this specific case, the model implies that: $\Box(\forall x x::\text{Heart} \rightarrow (\mathcal{E}(x) \rightarrow \exists y y::\text{Body} \wedge d_1(x, \text{Heart}, y, \text{Body})))$.

Now, suppose that we have a model such as the one represented of figure 7.19.

Figure 7-19 Example of direct functional parts of type 1



In this case, both the relationships between Heart and Body, and between Mitral Valve and Heart, are mapped in the instance level to cases of *direct functional parthood(1)*, i.e.,

$$(i) \Box(\forall x x::\text{Heart} \rightarrow (\mathcal{E}(x) \rightarrow \exists y y::\text{Body} \wedge d_1(x, \text{Heart}, y, \text{Body})))$$

⁶⁸ See definition of a functional complex in chapter 5.

$$(ii) \quad \Box(\forall x x::\text{MitralValve} \rightarrow (\mathcal{E}(x) \rightarrow \exists y y::\text{Heart} \\ \wedge d_1(x, \text{MitralValve}, y, \text{Heart})))$$

The important question at this point is: from (i) and (ii), can we derive formula (iii) below?

$$(iii) \quad \Box(\forall x x::\text{MitralValve} \rightarrow (\mathcal{E}(x) \rightarrow \exists y y::\text{Body} \\ \wedge d_1(x, \text{MitralValve}, y, \text{Body})))$$

We here adopt the so-called *mereological continuism* (Simons, 1987), which advocates that parthood only holds between existents. Since d_1 implies parthood, we conclude that d_1 also holds exclusively between existents. Thus, if in every world w that $x::X$ exists there is a $y::Y$ such $d_1(x, X, y, Y)$ in w then in every world w in which $x::X$ exists there is a $y::Y$ which also exists in w . Additionally, if in every world w that $y::Y$ exists there is a $z::Z$ such that $d_1(y, Y, z, Z)$ in w then in every world w in which $y::Y$ exists there is a $z::Z$ which also exists in w . From this we conclude that in every world w that $x::X$ exists there are $y::Y$ and $z::Z$ that also exist in w and both $d_1(x, X, y, Y)$ and $d_1(y, Y, z, Z)$ are true in that world. Consequently, we have that (iii) follows from (i) and (ii) iff d_1 is transitive. Thus, the question above can be rephrased as: is direct functional parthood(1) a transitive relation?

In the sequel we demonstrate that this is indeed the case. The abbreviations in the proofs are: (a) TFP (transitivity of formal parthood); (b) TLI (transitivity of the logical implication); (c) EC (Elimination of the Disjunction), and (d) IC (Introduction of the Disjunction).

(T1) Theorem 1: $d_1(x, X, y, Y) \wedge d_1(y, Y, z, Z) \rightarrow d_1(x, X, z, Z)$

Proof:

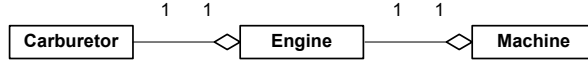
- | | |
|---|-------------------|
| 1. $d_1(x, X, y, Y)$ | T1 |
| 2. $d_1(y, Y, z, Z)$ | T1 |
| 3. $(x < y) \wedge \text{IFD}(x, X, y, Y)$ | 1, Definition 7.1 |
| 4. $\text{GFD}(X, Y) \wedge x::X \wedge y::Y \wedge (F(x, X) \rightarrow F(y, Y))$ | 3, (8) |
| 5. $(y < z) \wedge \text{IFD}(y, Y, z, Z)$ | 2, Definition 7.1 |
| 6. $\text{GFD}(Y, Z) \wedge y::Y \wedge z::Z \wedge (F(y, Y) \rightarrow F(z, Z))$ | 5, (8) |
| 7. $(x < z)$ | 3, 5, TFP |
| 8. $\text{GFD}(X, Z)$ | 4, 6, (7) |
| 9. $(x::X) \wedge (z::Z)$ | 4, 6, EC |
| 10. $(F(x, X) \rightarrow F(z, Z))$ | 4, 6, TLI |
| 11. $\text{GFD}(X, Z) \wedge (x::X) \wedge (z::Z) \wedge (F(x, X) \rightarrow F(z, Z))$ | 8, 9, 10, IC |

- | | |
|---|--------------------|
| 12. $\text{IFD}(x, X, z, Z)$ | 11, (8) |
| 13. $(x < z) \wedge \text{IFD}(x, X, z, Z)$ | 7, 12, IC |
| 14. $d_1(x, X, z, Z)$ | 13, Definition 7.1 |

□

We can generalize this result for any chain of direct functional dependence in a model. Another example of such case is depicted in figure 7.20.

Figure 7-20 Examples of direct functional parts of type 1; transitivity always hold across parthood relations of this type



In models such 7.13 and 7.20, the mandatory parthood relation represents functional dependence in both directions. Take for instance figure 7.20. As discussed in chapter 5, the minimum cardinality constraint of 1 in the Engine association end of the aggregation relation between Carburetor and Engine implies that every instance of Carburetor necessitates an Engine to function as a Carburetor. Likewise, the minimum cardinality constraint of 1 in the Carburetor association end of that relation implies that every Engine necessitates a Carburetor to function as an Engine. (Vieu & Aurnague, 2005) name this type of functional parthood in which x is part of y but y as Y is individually functionally dependent on x as an X direct functional parthood (2):

Definition 7.2 (Direct Functional Part of type 2): An individual x instance of X is a direct functional part of type 2 of an individual y of type Y (symbolized as $d_2(x, X, y, Y)$) iff x is a part of y and y is individually functionally dependent of x . Formally, $d_2(x, X, y, Y) \equiv (x < y) \wedge \text{IFD}(y, Y, x, X)$.

■

Examples of d_2 include wall-house, engine-car, electron-atom, atom-molecule, finger-hand, hand-arm, cell-heart, and feather-canary.

In the sequel, we prove that d_2 is also transitive.

(T2) Theorem 2: $d_2(x, X, y, Y) \wedge d_2(y, Y, z, Z) \rightarrow d_2(x, X, z, Z)$

Proof:

- | | |
|--|-------------------|
| (1). $d_2(x, X, y, Y)$ | T2 |
| (2). $d_2(y, Y, z, Z)$ | T2 |
| (3). $(x < y) \wedge \text{IFD}(y, Y, x, X)$ | 1, Definition 7.2 |
| (4). $\text{GFD}(Y, X) \wedge y::Y \wedge x::X \wedge (F(y, Y) \rightarrow F(x, X))$ | 3, (8) |

- | | |
|--|--------------------|
| (5). $(y < z) \wedge \text{IFD}(z, Z, y, Y)$ | 2, Definition 7.2 |
| (6). $\text{GFD}(Z, Y) \wedge z::Z \wedge y::Y \wedge (F(z, Z) \rightarrow F(y, Y))$ | 5, (8) |
| (7). $(x < z)$ | 3, 5, TFP |
| (8). $\text{GFD}(Z, X)$ | 4, 6, (7) |
| (9). $(z::Z) \wedge (x::X)$ | 4, 6, EC |
| (10). $(F(z, Z) \rightarrow F(x, X))$ | 4, 6, TLI |
| (11). $\text{GFD}(Z, X) \wedge (z::Z) \wedge (x::X) (F(z, Z) \rightarrow F(x, X))$ | 8, 9, 10, IC |
| (12). $\text{IFD}(z, Z, x, X)$ | 11, (8) |
| (13). $(x < z) \wedge \text{IFD}(z, Z, x, X)$ | 7, 12, IC |
| (14). $d_2(x, X, z, Z)$ | 13, Definition 7.2 |

□

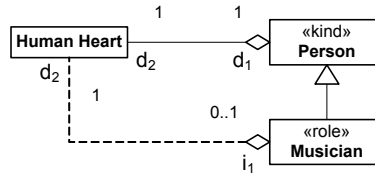
Whenever in a conceptual model we have a representation of a mandatory parthood relation between complex objects such as in figures 7.13 and 7.20, we have both a case of d_1 and a case of d_2 . In particular, the model of figure 7.20 implies both formulas below:

- (i) $\square(\forall x x::\text{Carburator} \rightarrow (\exists(x) \rightarrow \exists y y::\text{Engine} \wedge d_1(x, \text{Carburator}, y, \text{Engine})))$
- (ii) $\square(\forall x x::\text{Engine} \rightarrow (\exists(x) \rightarrow \exists y y::\text{Carburator} \wedge d_2(y, \text{Carburator}, x, \text{Engine})))$

Since both d_1 and d_2 are transitive, we maintain that transitivity holds within any chain of direct functional dependence relations in a conceptual model.

Now, take for instance the relationship depicted in figure 7.21 below.

Figure 7-21 Example of an indirect functional parthood of type 1 (from Human Heart to Musician)



Every human heart necessitates a person, and every person necessitates a human heart, i.e., both d_1 and d_2 hold between direct instances of human heart and person. Moreover, every musician is a person. So, as any person, a musician necessitates a human heart, i.e., d_2 holds also between instances of human heart and musician. However, it is not the case that a *direct functional dependence* holds between human heart and musician. A human heart necessitates a person, but this person does not have to be a musician (this is made evident by the cardinality 0..1 of the inherited relation between these two universals). This type of relationship is termed *indirect*

functional parthood (1) in (Vieu & Aurnague, 2005) and it is defined as follows:

Definition 7.3 (Indirect Functional Part of type 1): $i_1(x, X, y, Y) \equiv (x < y) \wedge \text{IIFD}(x, X, y, Y)$. $\text{IIFD}(x, X, y, Y)$ is the relation of *individual indirect functional dependence* and is defined as

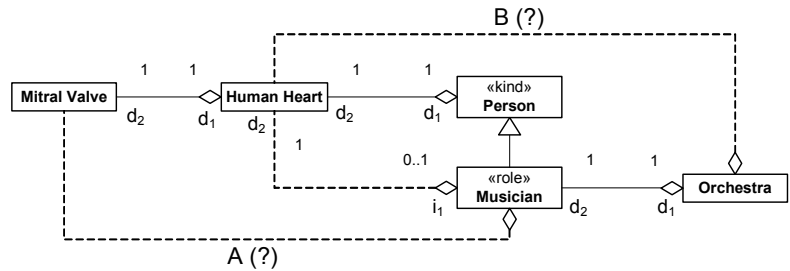
$$(9). \text{IIFD}(x, X, y, Y) \equiv y :: Y \wedge \exists Z (\text{Subtype}(Y, Z) \wedge \text{IFD}(x, X, y, Z)).$$

■

To put it in a simple way, x as an X is individually indirect functional dependent of y as a Y iff for x to function as an X , y must function as a Z , whereas Z is a more general universal (subsuming that Y) that y instantiates. Examples of i_1 include handle-door (with “movable entity” for type subsuming “door”), door-house (with “wall, enclosure or building” subsuming “house”), engine-car (with “machine” subsuming “car”), brick-wall (with “construction” subsuming “wall”), valve-carburetor (with “fluid-holding device” subsuming carburetor), cell-heart (with “organ” subsuming “heart”), feather-canary (with “bird” subsuming “canary”).

Now, take the model depicted in figure 7.22 below. There are two potential parthood relations **A** and **B**. The relation **A** between Mitral Valve and Musician holds iff transitivity holds across (Mitral Valve $\xrightarrow{d1}$ Human Heart) and (Human Heart $\xrightarrow{i1}$ Musician), since in the other reading of these relations, i.e., (Mitral Valve $\xrightarrow{d2}$ Human Heart) and (Human Heart $\xrightarrow{d1}$ Musician), transitivity is already guaranteed by theorem (T2). To put it baldly, relation **A** is transitive in this case iff $d_1(x, X, y, Y) \wedge i_1(y, Y, z, Z) \rightarrow i_1(x, X, z, Z)$ is a theorem. Likewise, relation **B** is transitive in this case iff $i_1(x, X, y, Y) \wedge d_1(y, Y, z, Z) \rightarrow i_1(x, X, z, Z)$ is a theorem.

Figure 7-22 Two candidate parthood relations due to transitivity.



As we show in the sequel, $d_1(x, X, y, Y) \wedge i_1(y, Y, z, Z) \rightarrow d_1(x, X, z, Z) \vee i_1(x, X, z, Z)$ is a theorem (T3) while $i_1(x, X, y, Y) \wedge d_1(y, Y, z, Z) \rightarrow d_1(x, X, z, Z) \vee i_1(x, X, z, Z)$ is not. Therefore, whilst **A** is a case of indirect functional parthood between Mitral Valve and Musician, relation **B** is not warranted and, hence, must not exist in figure 7.22.

(T3) Theorem 3: $d_1(x, X, y, Y) \wedge i_1(y, Y, z, Z) \rightarrow i_1(x, X, z, Z)$

Proof:

- | | |
|--|--------------------|
| (1). $d_1(x, X, y, Y)$ | T3 |
| (2). $i_1(y, Y, z, Z)$ | T3 |
| (3). $(x < y) \wedge \text{IFD}(x, X, y, Y)$ | 1, Definition 7.1 |
| (4). $\text{GFD}(X, Y) \wedge x::X \wedge y::Y \wedge (F(x, X) \rightarrow F(y, Y))$ | 3, (8) |
| (5). $(y < z) \wedge \text{IIFD}(y, Y, z, Z)$ | 2, Definition 7.3 |
| (6). $z::Z \wedge \exists W (\text{Subtype}(Z, W) \wedge \text{IFD}(y, Y, z, W))$ | 5, (9) |
| (7). $\text{GFD}(Y, W) \wedge y::Y \wedge z::W \wedge (F(y, Y) \rightarrow F(z, W))$ | 6, (8) |
| (8). $(x < z)$ | 3,5,TFP |
| (9). $\text{GFD}(X, W)$ | 4,7, (7) |
| (10). $(x::X) \wedge (z::W)$ | 4,7, EC |
| (11). $(F(x, X) \rightarrow F(z, W))$ | 4,7,TLI |
| (12). $\text{GFD}(X, W) \wedge (x::X) \wedge (z::W) (F(x, X) \rightarrow F(z, W))$ | 9,10,11,IC |
| (13). $\text{IFD}(x, X, z, W)$ | 12, (8) |
| (14). $z::Z \wedge \exists W (\text{Subtype}(Z, W) \wedge \text{IFD}(x, X, z, W))$ | 6,13,IC |
| (15). $\text{IIFD}(x, X, z, Z)$ | 14, (9) |
| (16). $(x < z) \wedge \text{IIFD}(x, X, z, Z)$ | 8,15, IC |
| (17). $i_1(x, X, z, Z)$ | 16, Definition 7.3 |

□

Let us now modify the model of figure 7.22 to depict a more realistic conceptualization. In this modified specification (figure 7.23) we have that every Blood Pump is part of a Circulatory System and necessitates a Circulatory System in order to work as such (d_1). Likewise, every Circulatory System has as part a Blood Pump and necessitates the latter to work as such (d_2). As any Blood Pump, a Biological Heart is part of a Circulatory System and necessitates a Circulatory System to work as such, i.e., *direct functional dependence* (2) is inherited by Biological Heart from the subsuming universal. The same obviously holds for Artificial Heart. However, it is not the case that a Circulatory System is directly functionally dependent of a Biological Heart specifically. To put it in an alternative way, a Circulatory System, in order to function as such, relies on the behavior of

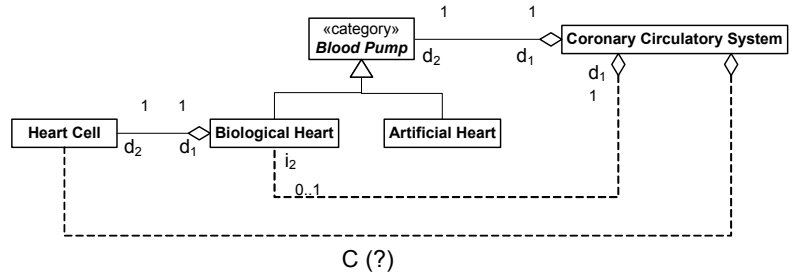
a Blood Pump, but this behavior does not have to be afforded in the specific way a Biological Heart does. In (Vieu & Aurnague, 2005), this type of relationship between Biological Heart and Circulatory System is termed *indirect functional parthood* (2) and it is defined as follows:

Definition 7.4 (Indirect Functional Part of type 2): $i_2(x, X, y, Y) \equiv (x < y) \wedge \text{IIFD}(y, Y, x, X)$.

■

Examples of i_2 include heart-circulatory system (with “blood pump” subsuming “heart”), brick-wall (with “construction material” subsuming “brick”).

Figure 7-23 Example of an indirect functional parthood of type 2 (from Biological Heart to Coronary Circulatory System) and of a candidate parthood relationship (C) due to transitivity



Once more, we have the question: does transitivity hold across (Heart Cell $\xrightarrow{d2}$ Biological Heart) and (Biological Heart $\xrightarrow{i2}$ Coronary Circulatory System)? In the other reading we have (Heart Cell Valve $\xrightarrow{d1}$ Biological Heart) and (Biological Heart $\xrightarrow{d1}$ Coronary Circulatory System), thus, relation **C** is warranted iff the question above is answered affirmatively. The answer in this case is negative, since $d_2(x, X, y, Y) \wedge i_2(y, Y, z, Z) \rightarrow d_2(x, X, z, Z) \vee i_2(x, X, z, Z)$ cannot be shown to be a theorem in this theory. However, the following is a theorem:

(T4) Theorem 4: $i_2(x, X, y, Y) \wedge d_2(y, Y, z, Z) \rightarrow i_2(x, X, z, Z)$

Proof:

- | | |
|---|-------------------|
| (15). $i_2(x, X, y, Y)$ | T4 |
| (16). $d_2(y, Y, z, Z)$ | T4 |
| (17). $(x < y) \wedge \text{IIFD}(y, Y, x, X)$ | 1, Definition 7.4 |
| (18). $(y < z) \wedge \text{IFD}(z, Z, y, Y)$ | 2, Definition 7.2 |
| (19). $x::X \wedge \exists W (\text{Subtype}(X, W) \wedge \text{IFD}(y, Y, x, W))$ | 3, (9) |
| (20). $\text{GFD}(Y, W) \wedge y::Y \wedge x::W \wedge (F(y, Y) \rightarrow F(x, W))$ | 5, (8) |

- (21). $GFD(Z,Y) \wedge z::Z \wedge y::Y \wedge (F(z,Z) \rightarrow F(y,Y))$ 4, (8)
 (22). $(x < z)$ 3,4, TFP
 (23). $GFD(Z,W)$ 6,7, (7)
 (24). $(z::Z) \wedge (x::W)$ 6,7,EC
 (25). $(F(z,Z) \rightarrow F(x,W))$ 6,7,TLI
 (26). $GFD(Z,W) \wedge (z::Z) \wedge (x::W) (F(z,Z) \rightarrow F(x,W))$ 9,10,11,IC
 (27). $IFD(z,Z,x,W)$ 12,(8)
 (28). $x::X \wedge \exists W (Subtype(X,W) \wedge IFD(z,Z,x,W))$ 5,13,IC
 (29). $IIFD(z,Z,x,X)$ 14,(9)
 (30). $(x < z) \wedge IIFD(z,Z,x,X)$ 8,15,IC
 (31). $i_2(x,X,z,Z)$ 16, Definition 7.4

□

Due to this theorem we have that the relation D between Biological Heart and Circulatory System (depicted in figure 7.24 below) is warranted, since transitivity holds across (Biological Heart $\xrightarrow{i_2}$ Coronary Circulatory System) and (Coronary Circulatory System $\xrightarrow{d_2}$ Circulatory System) in this case.

Figure 7-24 Example of an indirect functional parthood of type 2 due to transitivity (from Biological Heart to Circulatory System)

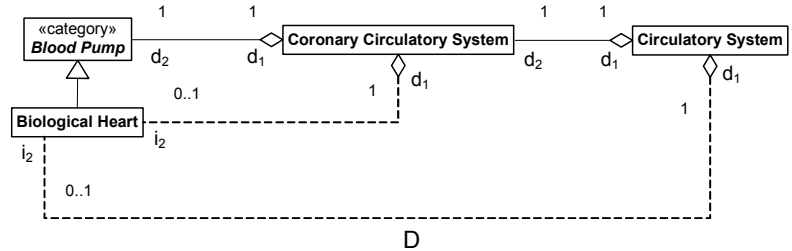
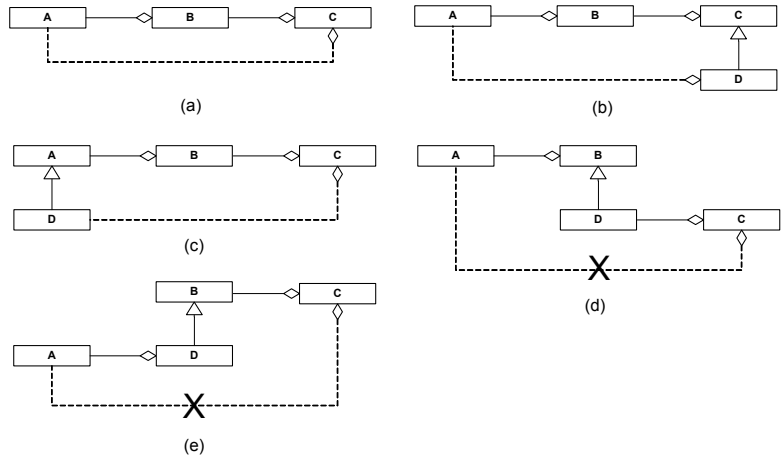


Figure 7.25 below summarizes the results of this section in the following manner: Mandatory parthood between complex objects (functional parthood) is a non-transitive relation (i.e., transitive in certain cases and intransitive in others). Transitivity can be guaranteed for these relations only in cases where the patterns of figures (7.25.a-c) occur.

Figure 7-25 The patterns of figures (a), (b) and (c) represent cases in which a derived transitive parthood relation can be inferred. Intransitive cases are shown in figures (d) and (e)



7.5 Final Considerations

In this chapter, we build upon some of the results developed in chapters 4, 5 and 6 of this thesis to provide a foundation for the modeling concept of roles.

First, by employing the categories from the theory of universals presented in chapter 4 and its postulates, we were able to propose a design pattern to target a recurrent problem in role modeling discussed in the literature. We believe that the definition of design patterns capturing standard solutions for ontological modeling problems contribute greatly to the task of defining sound engineering tools for conceptual modeling. Nonetheless, the investigation of ontological design patterns is still in its infancy, and very few examples exist in the literature. Two examples are the *whole-part* design pattern introduced in (Guizzardi & Falbo & Pereira Filho, 2002) and the *Inflammation* pattern proposed in (Gangemi & Catenacci & Battaglia, 2004).

In chapter 5, we present a number of *modal* meta-properties characterizing part-whole relations. Here, we use these meta-properties to investigate their intertwining with anti-rigid universals and, in particular, with role universals standing as wholes in parthood relationships. We use the distinction between *de re* and *de dicto* modality in the literature of philosophical logic to characterize the different formal properties of the relations of specific dependence parthood from the whole to the part, depending whether the whole universal is rigid or anti-rigid.

By using the concept of role (as anti-rigid and relationally dependent sortals) introduced in chapter 4, and the concept of *qua individuals*

(externally dependent modes) introduced in chapter 6, we manage to reconcile two incompatible views of roles present in the literature, as well as to present a modeling solution that retains the benefits of both approaches.

Finally, in chapter 5, we have discussed the so-called problem of fallacious transitivity in part-whole relations. In that chapter, we address this problem for three types of conceptual parthood, namely, the *subQuantityOf*, *subCollectionOf* and *memberOf* relations. However, the remaining relation (*componentOf*) is the one which is most commonly represented in conceptual models, since most individuals represented in these models are functional complexes. As discussed in that chapter, transitivity among *componentOf* relations only holds in certain contexts, but the definition of these contexts typically demands a substantial knowledge of the modeling domain and of the characterizing relations that unify its entities. With the intent to provide some methodological tools for helping the modeler in this task, we propose a number of visual patterns that can be used to delimit these contexts in conceptual class diagrams. The patterns are elaborated by using the concepts of relators, qua individuals and existential dependency relations discussed in chapter 6, but chiefly by building on the pioneering theory of transitivity of linguistic functional parthood relations proposed by (Vieu & Aurnague, 2005).

